Math 1200 Lecture 12

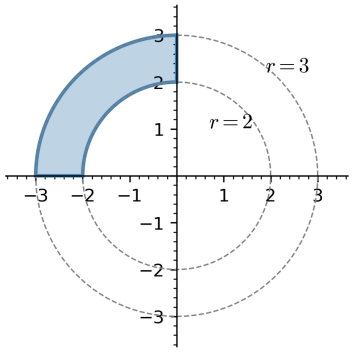
Oct 13 2026

Topics: 15.3: Double integrals in polar coordinates Name:_____
Collaborators:_____

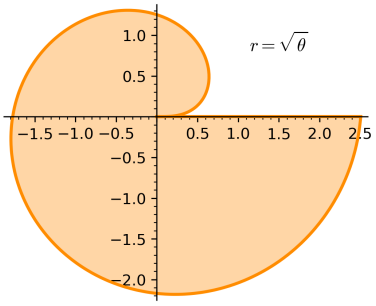
Group work 1 Evaluate the following double integrals

a) Calculate $\int_0^2 \int_0^{\sqrt{4-x^2}} \sqrt{x^2+y^2}\,dy\,dx$

b) Evaluate $\iint_R (2x-3y)\,dA$ where R is the region in the following figure



Group work 2 Find the area of the interior of the conch spiral traced by the polar curve $r = \sqrt{\theta}$ shown below.



Group work 3 Find the volume between the paraboloid $z = 1 - x^2 - y^2$ and the xy-plane.

Group work 4 The function $f(x,y) = e^{-x^2-y^2}$ is called the (2d) *Gaussian function* and it has many applications, especially in probability. Calculate $\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-x^2-y^2}dxdy$

Group work 5 (Challenge problem) From the previous calculation, deduce that

$$\int_{-\infty}^{\infty} e^{-x^2}dx = \sqrt{\pi}$$